

ROUSHAN KUMAR

AI Systems Engineer | LLM Orchestration & Developer Tooling Specialist

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PROFESSIONAL SUMMARY

AI Systems Engineer specializing in LLM application integration, real-time API gateways, vector search pipelines, and autonomous execution runtimes. Proficient in integrating Google Gemini, Cerebras (Llama-3.3), and Deepgram WebSocket APIs with Python (FastAPI) and vector databases (ChromaDB). Skilled at building self-healing code execution sandboxes, Tree-Sitter AST parser pipelines, and AI-assisted developer workflows with low latency.

TECHNICAL SKILLS

AI Engineering & APIs: Gemini API • Cerebras API • Deepgram API • LLM Orchestration • Prompt Engineering

Developer Tooling & Search: Tree-Sitter AST Compilers • ChromaDB (Vector Search) • Code Graph Indexing • Subprocess Isolation

Backend & Systems: Python (FastAPI) • Node.js • Express.js • C# (.NET Core) • REST APIs • WebSockets

Languages: C • Rust • Python • TypeScript • JavaScript • C# • Java • SQL

Databases & Caching: ChromaDB • Redis • PostgreSQL • MongoDB • SQLite • Supabase • Vector Embeddings

Testing & DevOps: PyTest • xUnit • JUnit 5 • Docker • CI/CD • GCP • Git • Vercel

TECHNICAL PROJECTS

Astra Vision - Developer Sandbox & Code Graph Parser (FastAPI, Python, Monaco, Tree-Sitter)

[Live Demo](#) | [GitHub](#)

- Product Architecture:** Architected an automated code-parsing platform and browser sandbox that allows developers to safely execute untrusted code while visualizing full-stack repository dependency graphs.
- Sandbox Runtime Challenge:** Built a self-healing Python execution engine with subprocess isolation and security checkpoints to intercept unauthorized OS file system calls and network socket requests in real time.
- Indexing Performance:** Integrated Tree-Sitter AST compilers and ChromaDB vector search to parse repository syntax nodes, enabling instant code graph dependency analysis under 200ms.

IDBI FinSync - AI-Powered Wealth & Financial Management Engine (Next.js, React, Fastify, Gemini API)

[Live Demo](#) | [GitHub](#)

- Product Architecture:** Co-created an intelligent personal financial management web app in a monorepo that unifies bank ledgers, investment portfolios, and expense streams into a live interactive dashboard.
- AI Integration Challenge:** Embedded "Mitra," an interactive AI wealth consultant leveraging Gemini API to analyze spending habits, detect budget anomalies, and deliver personalized financial guidance.
- Database & Concurrency:** Designed PostgreSQL transaction ledgers validated by Zod schemas and Fastify microservice endpoints, handling concurrent balance updates with zero data loss.

Alyra Lock - Secure Zero-Knowledge Password Vault (React, TypeScript, Express, MongoDB)

[Live Demo](#) | [GitHub](#)

- Product Architecture:** Engineered a client-side zero-knowledge password vault ensuring master encryption keys remain entirely isolated inside user browser memory, mitigating cloud database leak risks.
- Cryptographic Engineering:** Implemented Web Crypto API primitives utilizing AES-GCM (256-bit) payload encryption and PBKDF2 key derivation over 100,000 hashing iterations for key security.
- Performance & Security:** Developed high-throughput Express.js REST API sync endpoints with JWT authentication and strict CORS headers, keeping vault database sync latency under 200ms.

TECHNICAL SIMULATIONS & VIRTUAL EXPERIENCES

Commonwealth Bank (CommBank) - Software Engineering Virtual Simulation

June 2026

- Problem & Scope:** Diagnosed and resolved silent data overwrite bugs across high-traffic C#/.NET Core financial API controllers handling MongoDB document updates across financial services endpoints.
- Action Taken:** Extended C# Web API controllers with `$set` atomic operators for partial payload updates and modernized an interactive React/Redux Goal Manager UI component for banking clients.
- Engineering Outcome:** Authored comprehensive automated test suites using xUnit and Moq, covering complex boundary conditions to guarantee high reliability and transactional consistency.

Y Combinator (YC Startup - Shiptivity) - Software Engineering Virtual Simulation

June 2026

- Problem & Scope:** Fixed key constraint collisions and slow re-rendering performance on a real-time drag-and-drop Kanban workflow board used by engineering teams to track task state transitions.
- Action Taken:** Architected dynamic SQLite priority reordering logic using atomic transactions to update priority ranks sequentially and integrated Dragula drag-and-drop state hooks in React.
- Engineering Outcome:** Reduced UI component re-render cycles by 30% and patched OpenSSL compilation bottlenecks in legacy Node.js Webpack configurations to optimize runtime builds.

Walmart USA - Advanced Software Engineering Virtual Simulation

May 2026

- Problem & Scope:** Solved indexing overhead and performance bottlenecks in high-volume retail inventory processing queues handling unnormalized data ingestion from legacy spreadsheets.
- Action Taken:** Implemented a generic K-ary Max Heap structure in Java using fast bitwise shift operators (`<<`, `>>>`) and built 3NF database schemas with design pattern abstractions.
- Engineering Outcome:** Accelerated queue indexing calculations by 35% and engineered automated Python ETL pipelines using `csv` and `sqlite3` to clean and consolidate disparate spreadsheets.

EDUCATION

Bachelor of Computer Applications (BCA) - Amity University Noida

Expected July 2028

- Relevant Coursework:** Data Structures & Algorithms (DSA) • Database Management Systems (DBMS) • Operating Systems (OS) • Computer Networks • Object-Oriented Programming (OOP)